Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Switch and Router Configuration

Intro to VLANs and static routing. Create and configure VLANs on a switch.

Task: Configure a static route between two routers in a simulation.

- **1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.**

  - **Aim**
    - To create and configure two VLANs named **Sales** and **Support**, with VLAN IDs 10 and 20 respectively, on a switch using Cisco Packet Tracer.

  - **Introduction**
    - A Virtual Local Area Network (VLAN) is a logical division of a physical network into separate broadcast domains. VLANs help improve network organization, security, and traffic management.
    - In this practical, VLAN 10 is created for the Sales department and VLAN 20 is created for the Support department. These VLANs logically separate the two departments even though they use the same physical switch.

- **Requirements**
  - Cisco Packet Tracer software
  - One Cisco 2960 switch
  - A PC or laptop to access the switch CLI
  - Basic knowledge of Cisco IOS commands

- **VLAN Configuration Commands**
- Open Cisco Packet Tracer, place a Cisco 2960 switch in the workspace, click the switch, and open the **CLI** tab. Enter the following commands:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Support
Switch(config-vlan)# exit
Switch(config)# end
Switch# copy running-config startup-config
```

- Press **Enter** if the switch asks for the destination filename.

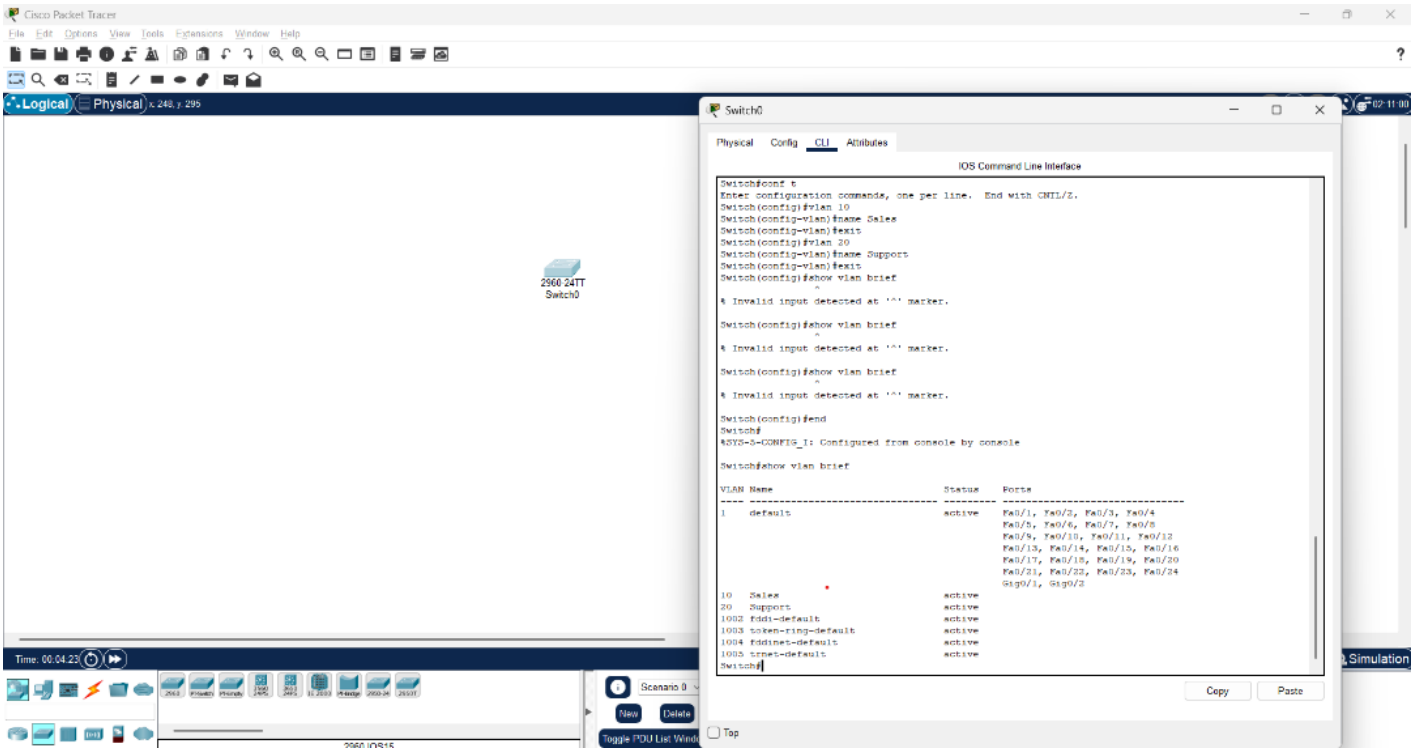
- **Verification Commands**
- To verify that both VLANs have been created, enter:

```
Switch# show vlan brief
```

- The output should list the following VLANs:

VLAN ID	VLAN Name	Status
10	Sales	Active
20	Support	Active

- The default VLAN 1 may also appear in the output.



(Creation and Verification of VLAN 10 (Sales) and VLAN 20 (Support) in Cisco Packet Tracer.)

➢ **Result**

➢ Two VLANs were configured on the Cisco switch:

- VLAN 10 — Sales
- VLAN 20 — Support

➢ The show vlan brief command can be used to verify their names, IDs, and status.

❖ **2. Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Use the interface range command to select multiple ports efficiently.**

➢ **Aim**

- To assign three switch ports to VLAN 10 (Sales) and two switch ports to VLAN 20 (Support) using the interface range command, and verify the VLAN membership using the show vlan brief command.

➢ **Introduction**

- A Virtual Local Area Network (VLAN) logically divides a network into separate broadcast domains. Switch ports can be assigned to different VLANs to separate departments and manage network traffic efficiently.
- In this practical, three switch ports are assigned to VLAN 10 named Sales, and two switch ports are assigned to VLAN 20 named Support.

➢ **Requirements**

- Cisco Packet Tracer

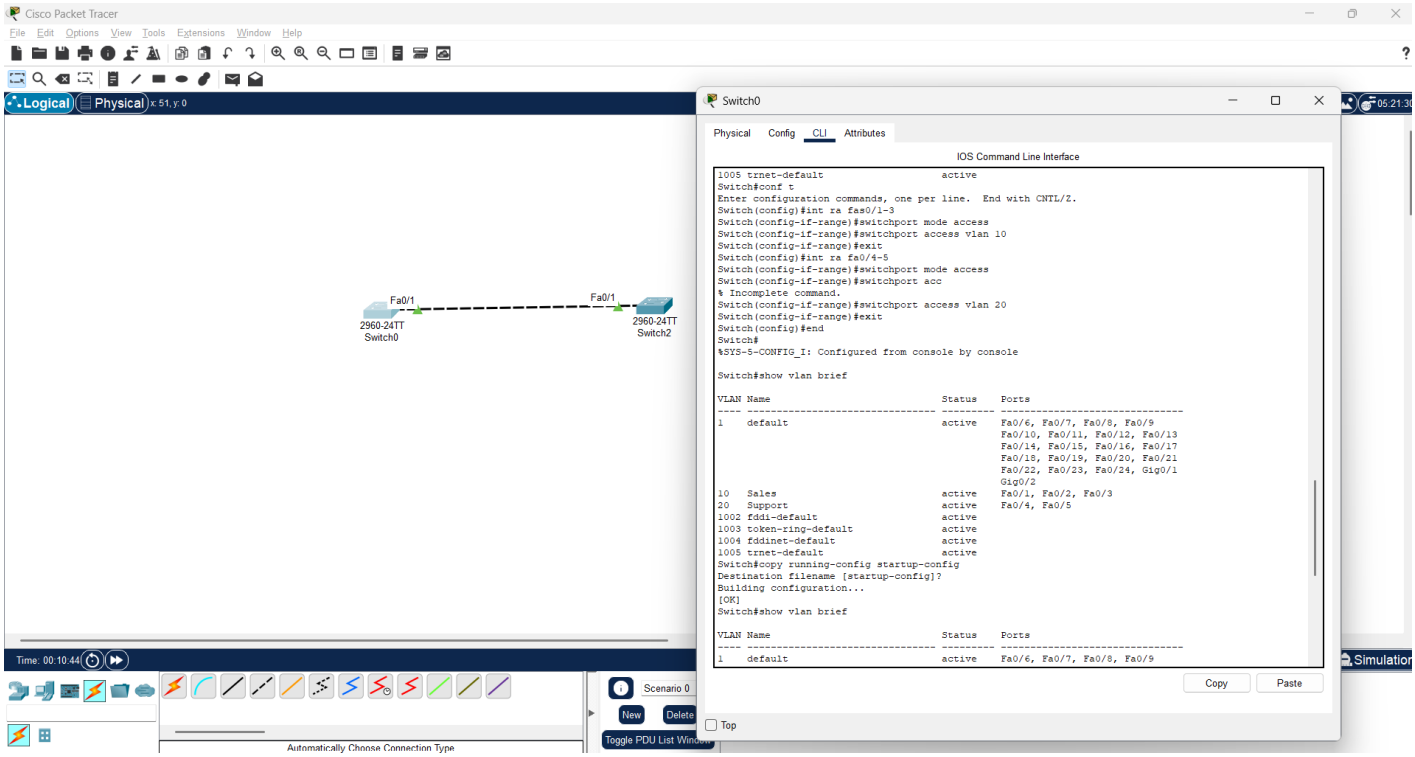
- One Cisco 2960 switch
- VLAN 10 (Sales) and VLAN 20 (Support), created on the switch

➢ **Configuration Procedure**
➢ Open the switch and select the **CLI** tab. Enter the following commands one by one.
➢ **Step 1: Enter privileged and configuration modes**
Switch> enable
Switch# configure terminal
➢ **Step 2: Assign three ports to VLAN 10 (Sales)**
➢ In this example, FastEthernet0/1 to FastEthernet0/3 are assigned to Sales.
Switch(config)# interface range fastEthernet 0/1-3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
➢ **Step 3: Assign two ports to VLAN 20 (Support)**
➢ In this example, FastEthernet0/4 and FastEthernet0/5 are assigned to Support.
Switch(config)# interface range fastEthernet 0/4-5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
➢ **Step 4: Save the configuration**
Switch(config)# end
Switch# copy running-config startup-config
➢ Press Enter if prompted to confirm the destination filename.

- **Verification**
- Enter the following command to verify the VLAN membership:
- Switch# show vlan brief
- The output should show the following port assignments:

VLAN ID	**VLAN Name**	**Assigned Switch Ports**
10	Sales	Fa0/1, Fa0/2, Fa0/3
20	Support	Fa0/4, Fa0/5

- The VLANs should be listed as active, and the specified ports should appear under their respective VLANs.



(Verification of Switch Port Membership in Sales and Support VLANs.)

- **Result**
- Three switch ports (FastEthernet0/1 to FastEthernet0/3) were assigned to VLAN 10 (Sales), and two switch ports (FastEthernet0/4 to FastEthernet0/5) were assigned to VLAN 20 (Support). The configuration was verified using the show vlan brief command.

❖ 3. Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

➢ Aim

- To connect two routers using an Ethernet link in Cisco Packet Tracer, assign IP addresses to their interfaces, configure a static route on Router1 to reach the LAN behind Router2, and test connectivity using the ping command.

➢ Requirements

- Cisco Packet Tracer
- Two routers
- Two switches
- Two PCs
- Ethernet cables

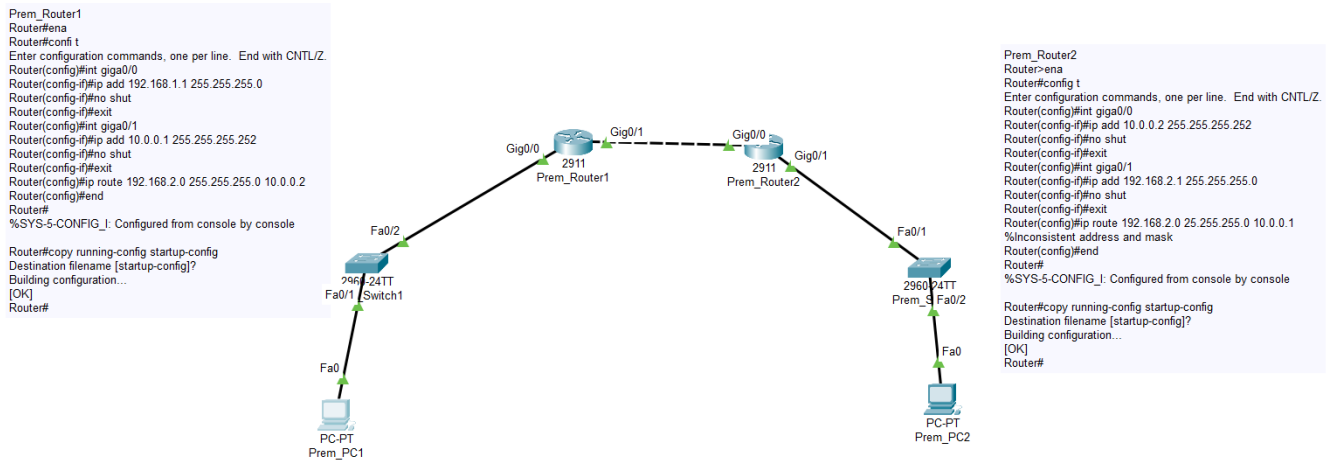
➢ IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask
Prem_PC1	FastEthernet0	192.168.1.10	255.255.255.0
Prem_Router1	GigabitEthernet0/0	192.168.1.1	255.255.255.0
Prem_Router1	GigabitEthernet0/1	10.0.0.1	255.255.255.252

Prem_Router2	GigabitEthernet0/0	10.0.0.2	255.255.255.252
Prem_Router2	GigabitEthernet0/1	192.168.2.1	255.255.255.0
Prem_PC2	FastEthernet0	192.168.2.10	255.255.255.0

- **Default Gateways:**
  - Prem_PC1: 192.168.1.1
  - Prem_PC2: 192.168.2.1

- **Procedure and Commands**
- **Step 1: Create the Network Topology**
- Place two routers, two switches and two PCs in Cisco Packet Tracer. Connect the PCs to the switches, the switches to the routers, and the two routers using an Ethernet link.

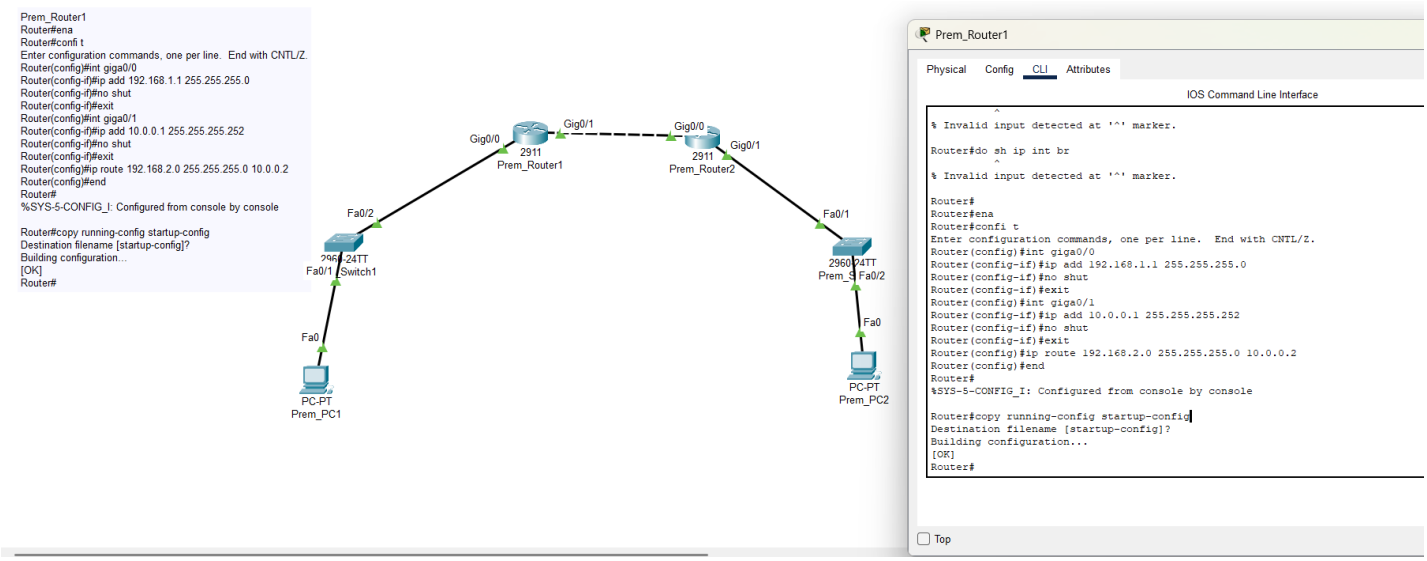


- **Step 2: Configure Router1**
- Open Prem_Router1 and enter the following commands:

  enable
  configure terminal
  interface gigabitEthernet 0/0
  ip address 192.168.1.1 255.255.255.0
  no shutdown
  exit
  interface gigabitEthernet 0/1
  ip address 10.0.0.1 255.255.255.252
  no shutdown
  exit
  ip route 192.168.2.0 255.255.255.0 10.0.0.2
  end

copy running-config startup-config

- The command ip route 192.168.2.0 255.255.255.0 10.0.0.2 configures a static route on Router1 to reach the LAN behind Router2.

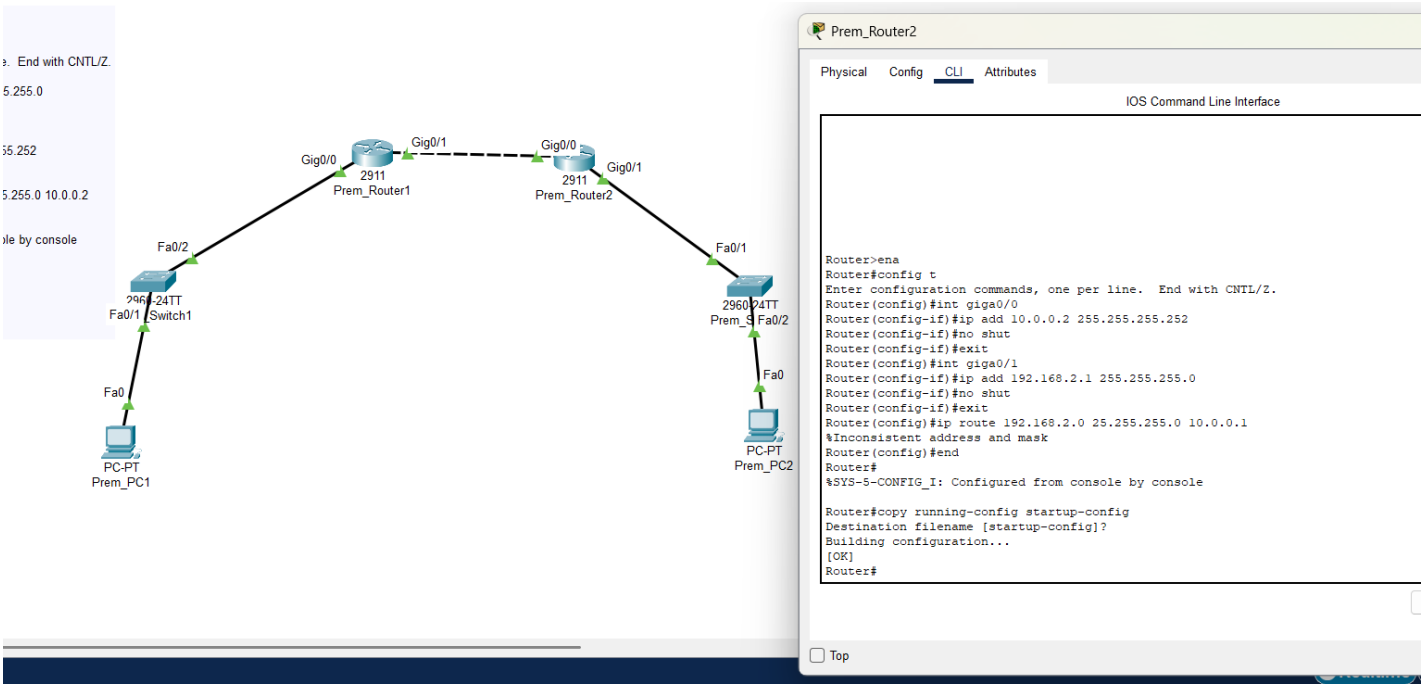


- **Step 3: Configure Router2**
- Open Prem_Router2 and enter the following commands:

enable
configure terminal
interface gigabitEthernet 0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

ip route 192.168.1.0 255.255.255.0 10.0.0.1
end
copy running-config startup-config

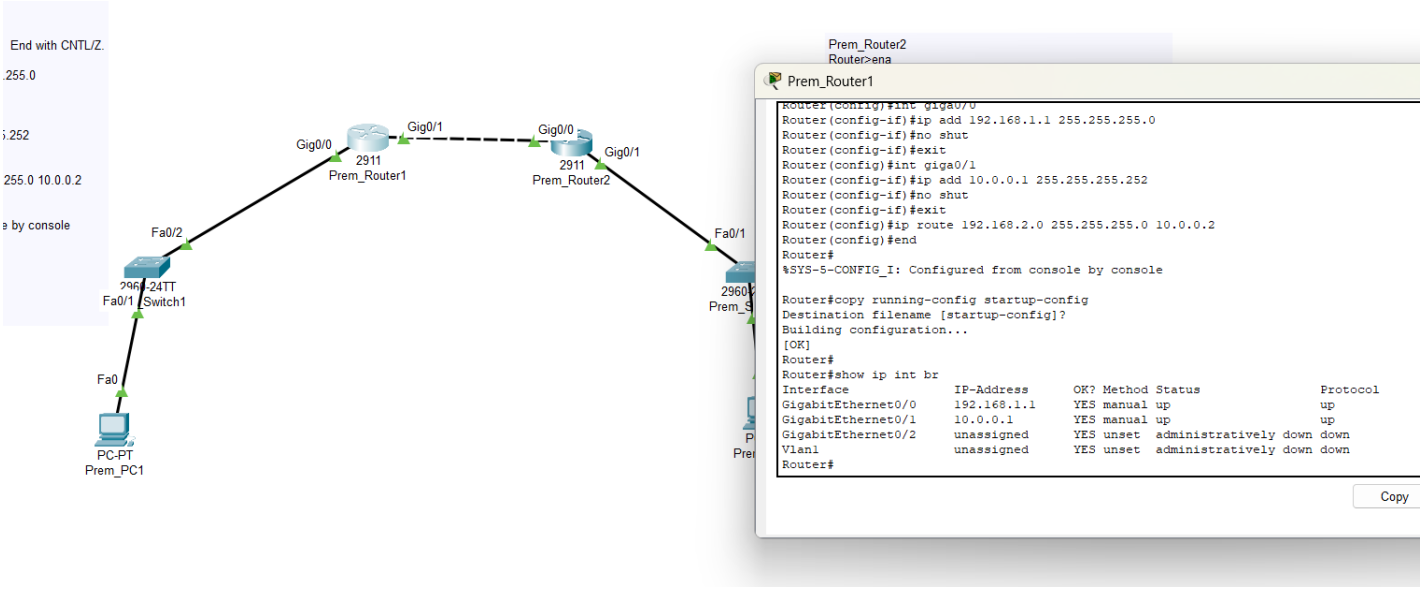
- The static route on Router2 provides the return path to the LAN behind Router1.



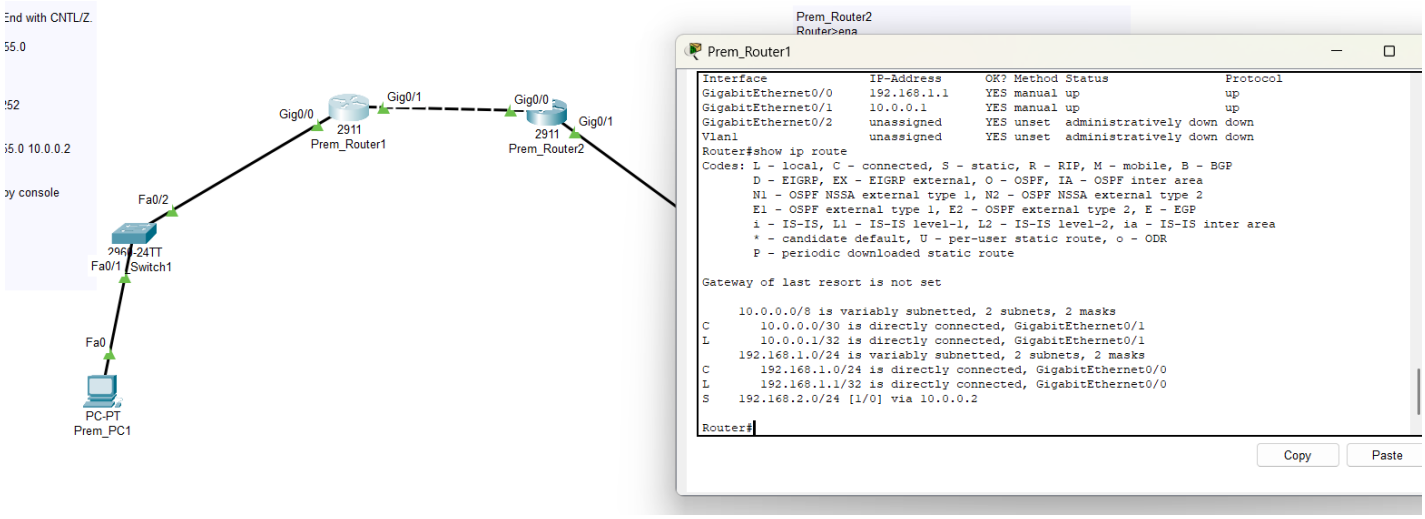
- **Step 4: Configure Prem_PC1 and Prem_PC2**
- Configure the PCs through **Desktop → IP Configuration**.
- **Prem_PC1:**
  IP Address: 192.168.1.10
  Subnet Mask: 255.255.255.0
  Default Gateway: 192.168.1.1
- **Prem_PC2:**

IP Address: 192.168.2.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

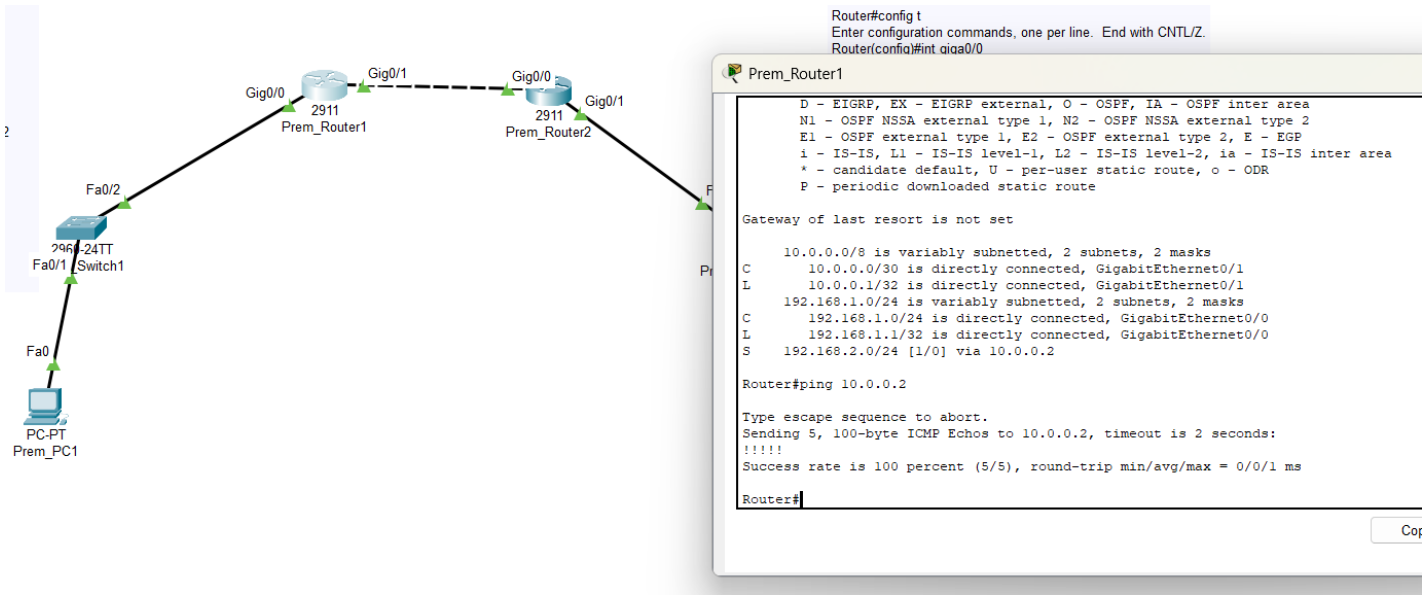
- **Step 5: Verify Router Interfaces**
- On the routers, use the following command:
  show ip interface brief
- This command displays the IP addresses and status of the router interfaces.

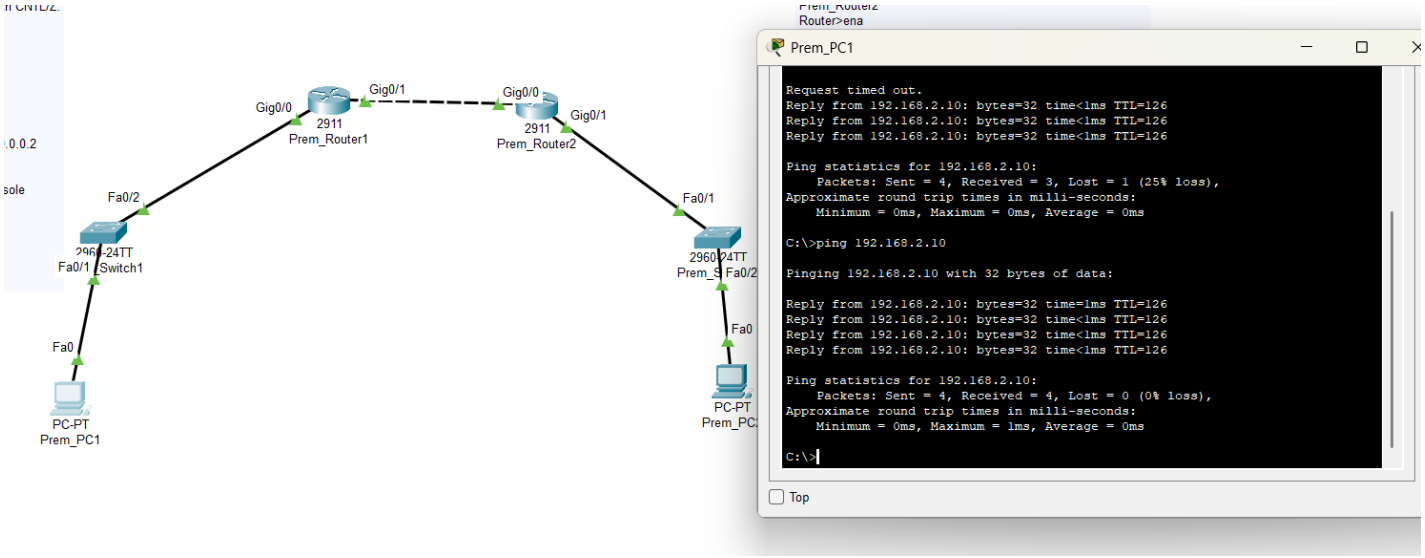


- **Step 6: Verify the Static Route on Router1**
- On Prem_Router1, enter:
  show ip route
- The routing table should show a static route similar to:
  S 192.168.2.0/24 [1/0] via 10.0.0.2



- **Step 7: Test Connectivity**
- On Prem_Router1, test the Ethernet connection to Router2:
  ping 10.0.0.2
- Next, open **Prem_PC1 → Desktop → Command Prompt** and enter:
  ping 192.168.2.10
- Successful replies confirm communication between PC1 and PC2 through the two routers. The first ping may time out while address resolution takes place, so repeat the command if necessary.





➢ Result

- Two routers were connected using an Ethernet link, IP addresses were assigned to their interfaces, and static routes were configured.

Connectivity between the two LANs was verified using the ping command.

❖ **4. Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication. Think about router-on-a-stick or Layer 3 switch concepts.**

➢ **Aim**

- To simulate communication between a device in the Sales VLAN and a device in the Support VLAN using Cisco Packet Tracer, understand why communication initially fails, and configure inter-VLAN routing using the router-on-a-stick method.

➢ **Introduction**

- A **VLAN (Virtual Local Area Network)** is a logical division of a network that separates devices into different broadcast domains, even when they are connected to the same physical switch.

➢ In this practical, two VLANs are created:

- **VLAN 10 – Sales**
- **VLAN 20 – Support**

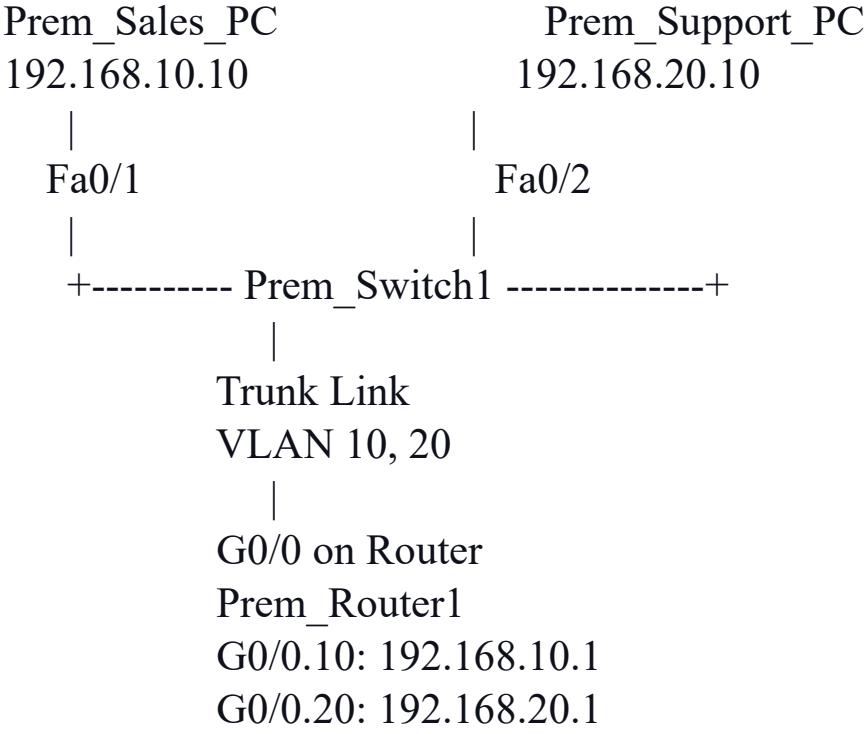
➢ A PC is connected to each VLAN. Initially, a PC in the Sales VLAN cannot communicate directly with a PC in the Support VLAN because they belong to different VLANs and different IP subnets.

➢ To allow communication between these VLANs, a router is configured using the **router-on-a-stick method**. The switch-to-router connection is

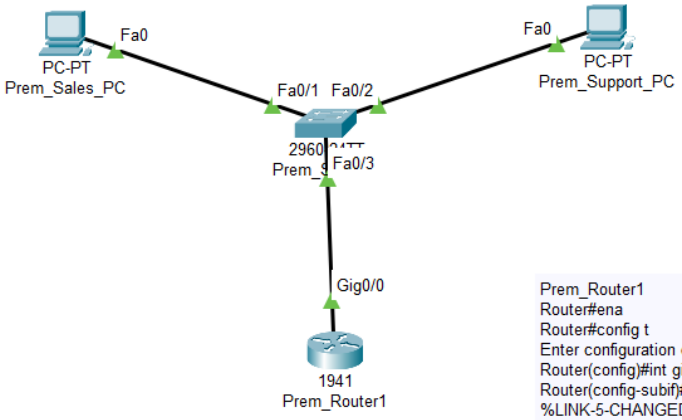
configured as a trunk, and the router uses separate subinterfaces for VLAN 10 and VLAN 20.

- **Requirements**
  1. Cisco Packet Tracer
  2. One router (Prem_Router1)
  3. One switch (Prem_Switch1)
  4. One Sales PC (Prem_Sales_PC)
  5. One Support PC (Prem_Support_PC)

- **Network Topology**
- The practical uses the following topology:



- The Sales PC belongs to VLAN 10, while the Support PC belongs to VLAN 20. The router provides communication between the two VLANs.




```
Prem_Switch
Switch>ena
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Support
Switch(config-vlan)#exit
Switch(config)#end
Switch(config)#int fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shut
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport acce
% Incomplete command.
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shut
```

```
Prem_Router1
Router#ena
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int giga0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip add 192.168.10.1 255.255.255.0
Router(config-subif)#no shut
Router(config-subif)#exit
Router(config)#int giga0/0.20
Router(config-subif)#ip add 192.168.20.1 255.255.255.0
Router(config-subif)#no shut
Router(config-subif)#end
Router#
```

- **IP Address Configuration**
- The following IP addresses are used in the practical.

Device	IP Address	Subnet Mask	Default Gateway
Prem_Sales_PC	192.168.10.10	255.255.255.0	192.168.10.1

Prem_Support_PC	192.168.20.10	255.255.255.0	192.168.20.1
Router VLAN 10 subinterface	192.168.10.1	255.255.255.0	Not applicable
Router VLAN 20 subinterface	192.168.20.1	255.255.255.0	Not applicable

- The default gateway of each PC is the router address assigned to its own VLAN.

- **Procedure**
- **Step 1: Create VLANs on the Switch**
- Open the CLI of Prem_Switch1 and enter the following commands:

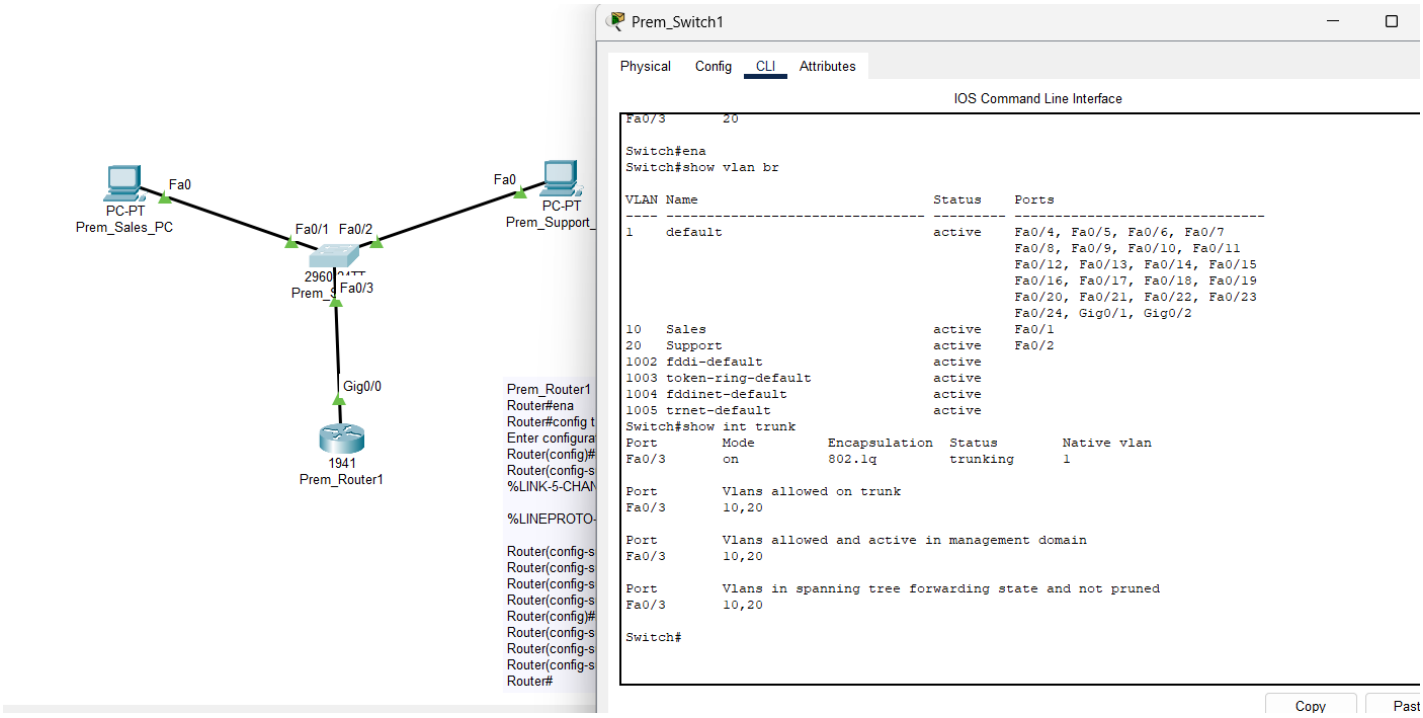
```
enable
configure terminal
vlan 10
name Sales
exit
vlan 20
name Support
exit
end
show vlan brief
```

- These commands create VLAN 10 named Sales and VLAN 20 named Support.

- **Step 2: Assign Switch Ports to VLANs**
- The Sales PC is connected to FastEthernet 0/1, and the Support PC is connected to FastEthernet 0/2.
- Enter the following commands on Prem_Switch1:

```
enable
configure terminal
interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
no shutdown
exit
interface fastEthernet 0/2
switchport mode access
switchport access vlan 20
no shutdown
exit
end
show vlan brief
```

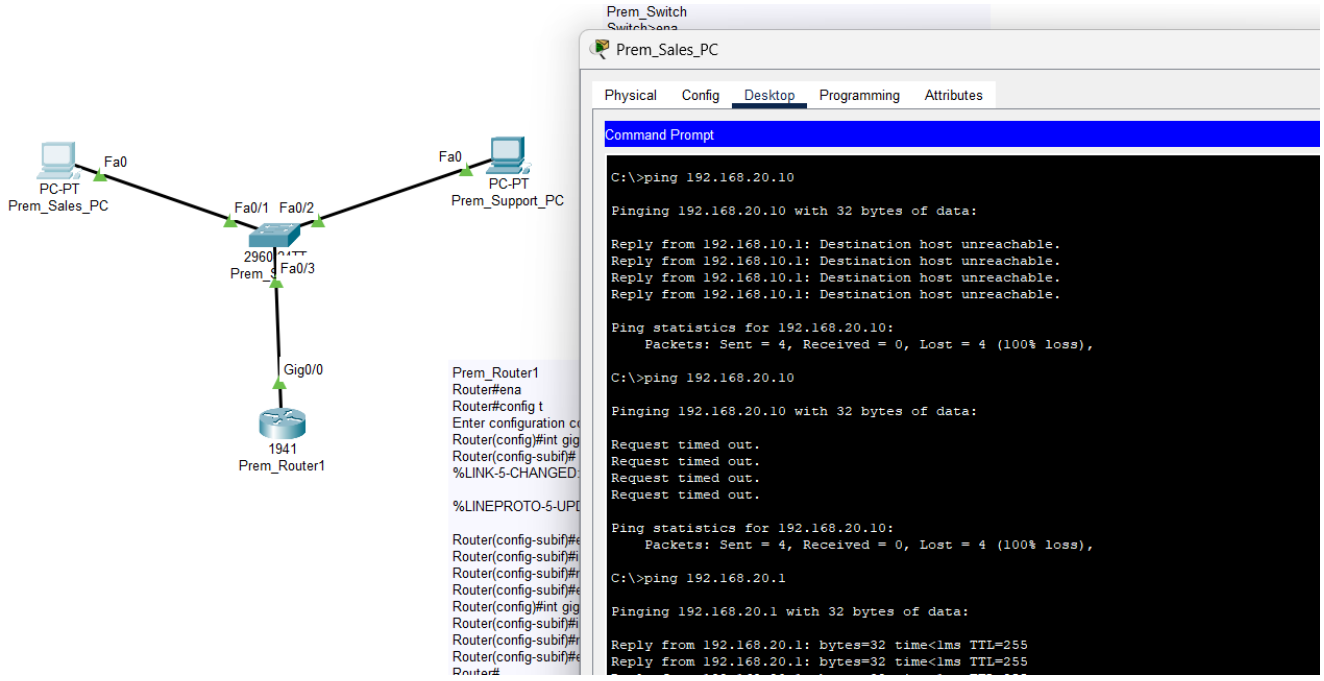
- These commands assign the Sales PC's switch port to VLAN 10 and the Support PC's switch port to VLAN 20.



- **Step 3: Configure IP Addresses on the PCs**
- Configure the following settings using **Desktop → IP Configuration** on each PC.
- **Sales PC**
  - IP Address: 192.168.10.10
  - Subnet Mask: 255.255.255.0
  - Default Gateway: 192.168.10.1
- **Support PC**
  - IP Address: 192.168.20.10
  - Subnet Mask: 255.255.255.0
  - Default Gateway: 192.168.20.1
- **Step 4: Test Communication Before Inter-VLAN Routing**
- Open **Desktop → Command Prompt** on Prem_Sales_PC and execute:

ping 192.168.20.10

- The ping initially failed because the PCs belonged to different VLANs and IP subnets, and inter-VLAN routing had not yet been configured.
- An initial test produced the following response:
  Reply from 192.168.10.1: Destination host unreachable.
- This indicated that the router was unable to forward the traffic to the Support PC at that stage of the configuration.



- **Step 5: Enable the Router Interface**
- Open the CLI of Prem_Router1 and enter:
  enable
  configure terminal
  interface gigabitEthernet 0/0

```
no shutdown
exit
```

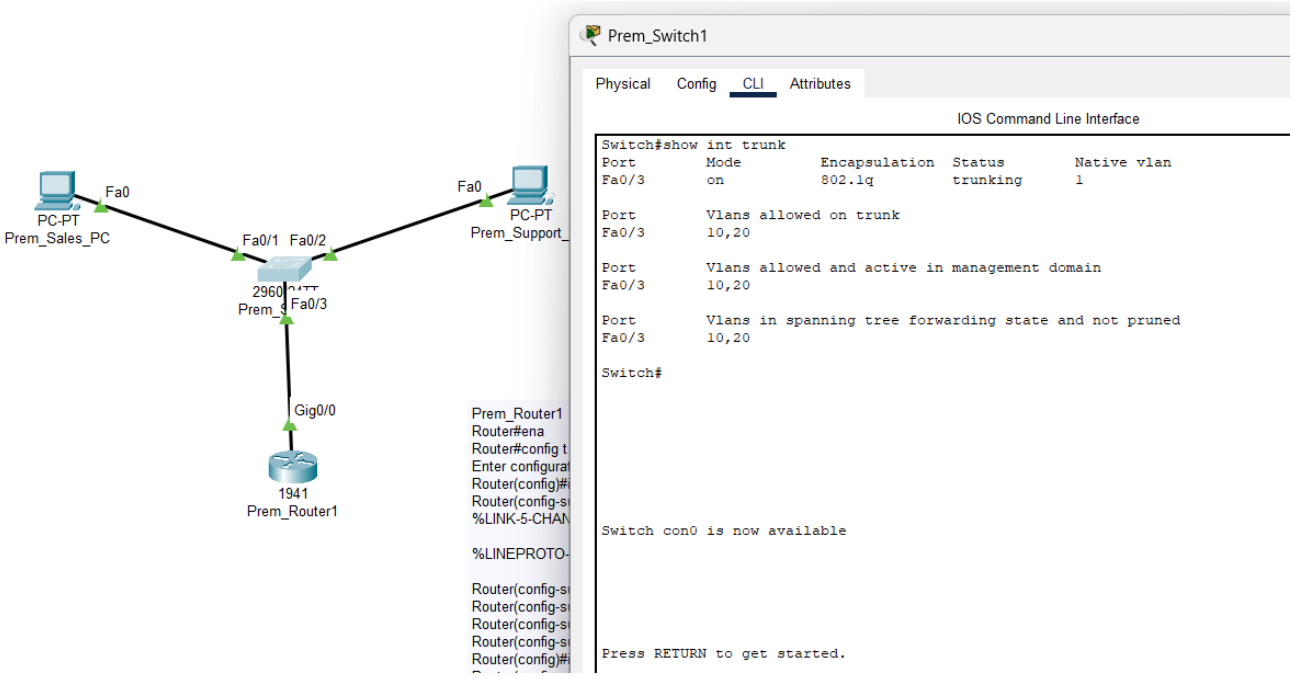
- This command enables the physical router interface connected to the switch.
- **Step 6: Configure Router Subinterfaces**
- Configure one subinterface for each VLAN on the router.

```
enable
configure terminal
interface gigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
end
```

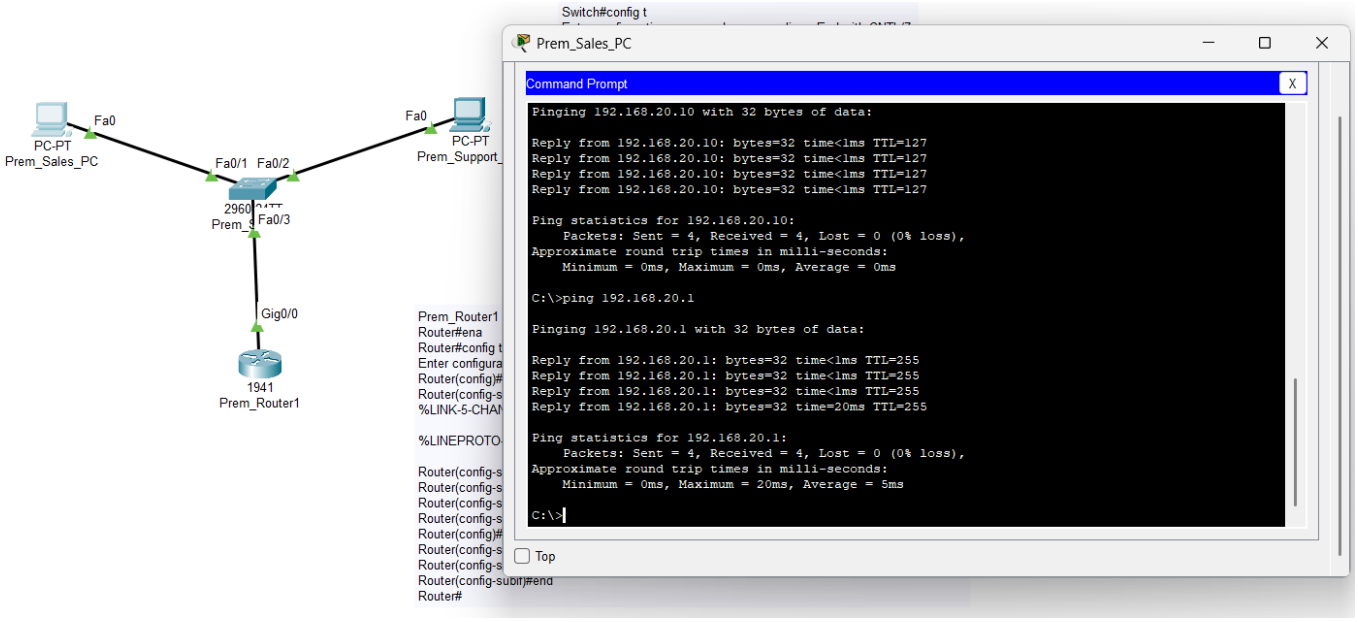
- The subinterface GigabitEthernet0/0.10 handles VLAN 10 traffic, while GigabitEthernet0/0.20 handles VLAN 20 traffic. The encapsulation dot1Q command identifies the VLAN associated with each subinterface.
- **Important correction made during the practical:** The initial address entered for the second subinterface was 192.198.20.1. It was corrected to 192.168.20.1 to match the Support subnet.
- **Step 7: Configure the Switch-to-Router Trunk**

- Configure the switch port connected to the router as a trunk. In this topology, the router's cable is connected to switch port FastEthernet 0/3.
- Enter the following commands on Prem_Switch1:

  enable
  configure terminal
  interface fastEthernet 0/3
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
  end
  show interfaces trunk
- The trunk carries traffic from VLAN 10 and VLAN 20 between the switch and router.



- **Step 8: Verify Router Interfaces**
- Use the following command on Prem_Router1:
  show ip interface brief
- The physical interface and both subinterfaces should show an operational status of up and protocol status of up.
- The relevant IP addresses should be:
  - GigabitEthernet0/0.10 – 192.168.10.1
  - GigabitEthernet0/0.20 – 192.168.20.1
- **Step 9: Test Inter-VLAN Communication Again**
- After correcting the router IP address, configuring the trunk, and setting the default gateways, execute the following command on Prem_Sales_PC:
  ping 192.168.20.10
- If the configuration is correct, the ping should receive replies from 192.168.20.10. If the first attempt times out, repeat the test after a few seconds.



- **Why Did the Initial Ping Fail?**
- The initial ping failed because the Sales PC and Support PC were placed in different VLANs and different IP subnets. A Layer 2 switch does not automatically route packets between VLANs.
- The router also requires correctly configured subinterfaces, VLAN tagging, a trunk connection, and appropriate default gateways. During this practical, the Support VLAN subinterface initially had an incorrect IP address, 192.198.20.1, which was corrected to 192.168.20.1.
- **How Was Inter-VLAN Communication Enabled?**
- Inter-VLAN communication was enabled using the **router-on-a-stick method**.
- The following configurations were necessary:
  1. VLAN 10 and VLAN 20 were created on the switch.
  2. The Sales and Support switch ports were assigned to their respective VLANs.
  3. The switch port connected to the router was configured as a trunk.
  4. Two router subinterfaces were created with 802.1Q VLAN tagging.
  5. Each subinterface was assigned an IP address to act as the default gateway for its VLAN.
  6. The correct default gateway was configured on each PC.
- A Layer 3 switch with VLAN interfaces (SVIs) and IP routing could also provide inter-VLAN communication, but the router-on-a-stick method was used in this practical.

- **Result**
- Two VLANs, Sales (VLAN 10) and Support (VLAN 20), were created in Cisco Packet Tracer. Router-on-a-stick configuration was performed to enable communication between the two VLANs.
- The initial ping failure demonstrated that devices in separate VLANs require Layer 3 routing for communication. After the configuration was corrected, inter-VLAN communication could be tested again using the ping command.